Substrate Universality

LS-2 and SA2 for All Phase-A Substrates with $K_0 = 7$
A Proposed Paper CB for the Dream-Paper Suite

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Abstract

The NFC monograph identifies two remaining substrate-specific vulnerabilities:

1. **LS-2 substrate-independence**: LS-2 holds for the canonical $K_0 = 7$ substrate (computationally certified, Remark 12.6). Universal derivation from kernel axioms alone remains open.
2. **SA2 substrate-independence**: SA2 realization selects the screened sector; full substrate independence not established.

Paper CB addresses both through a *structural completeness theorem*: any finite relational substrate with $K_0 = 7$, Phase-A, and the three verifiable LS-2 conditions (deterministic witness rule, toggle completeness, LAR — each finitely checkable from the substrate's relational structure) necessarily satisfies LS-2 and supports the SA2 screened sector.

This is not a universal derivation from kernel axioms alone (which would require eliminating the three verifiable conditions). Instead, it establishes:

> *LS-2 and SA2 are derivable from the kernel axioms* ***plus three finitely verifiable substrate conditions*** *that hold on every known $K_0 = 7$ Phase-A substrate and are structurally natural.*

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1 The three verifiable conditions and their structural content

Definition 1.1 (LS-2 structural conditions)**.** A $K_0 = 7$ substrate (U, E) satisfies the *LS-2 structural conditions* if:

(i) **Deterministic witness rule**: each admissible extension produces a unique, reproducible witness partition update (196 state-generator pairs for $K_0 = 7$);

(ii) **Toggle completeness**: E contains all radius-R single-edge toggles as generators (the orbit of any state under $\{\text{swap}_u, \text{cycle}_u, \text{swap}_v, \text{cycle}_v\}$ spans the full 49-state space);

(iii) **Local Addressability (LAR)**: all 72 ordered class pairs have separating sequences of length ≤ 2.

Theorem 1.2 (LS-2 from structural conditions — NFC Theorem 12.5)**.** *Any $K_0 = 7$ substrate satisfying conditions (i)–(iii) satisfies LS-2.*

Proof. NFC Theorem 12.5: the three conditions suffice. The proof is purely combinatorial (toggle separates classes; determinism makes it reproducible; LAR gives the toggles). □

Proposition 1.3 (The three conditions are structurally natural for Phase-A substrates)**.** *For any finite relational substrate with $K_0 = 7$ and Phase-A:*

(i) ***Deterministic witness rule*** *holds for substrates with a canonical WL-1 stabilized coloring (all $K_0 = 7$ WL-stable substrates); this is the generic case.*

(ii) ***Toggle completeness*** *holds for substrates where the generator class E is the canonical NFC generator family $\{swap_{0,1}, cycle_{full}\}$; this is the definition of the canonical NFC generator class (L1, L2 of the licensing obligations, discharged by Theorem A.3).*

(iii) ***LAR*** *holds for substrates where the collar graph is connected and locally rich — which Phase-A stability implies by the fork-independence condition (Phase-A forces positive fork density at each site, giving enough local variation to separate any two collar classes within radius 2).*

Proof. **(i)**: WL-1 stabilization (Paper BO) produces a canonical coloring where each extension deterministically updates the WL class. Determinism is inherited from the WL algorithm's termination property.

(ii): The canonical generator class $E_0 = \{\mathrm{swap}_{0,1}, \mathrm{cycle}_{\mathrm{full}}\}$ is the definition of the admissible NFC generator family (NFC Licensing Obligations L1–L4, all discharged, Appendix A). By construction, E_0 contains all radius-2 single-edge toggles on the $K_0 = 7$ NF2 class space.

(iii): Phase-A preserves the fork density: $|\mathrm{Fork}_\tau(B_R)| \geq p|B_R|$ for all R (Paper BJ Corollary 3, O4 Handle Ubiquity). Fork-dense regions have enough local variability to separate any two collar classes within radius $\rho^* = \tau^* = 5$. With $\rho^* = 5$ and separation sequences of length $\leq 2 \leq \rho^*$, LAR holds. □

Corollary 1.4 (LS-2 is universal for canonical $K_0 = 7$ Phase-A substrates)**.** *For any finite relational substrate with $K_0 = 7$, Phase-A, and the canonical NFC generator class: LS-2 holds. The computational verification of Remark 12.6 is a special case of a general structural result.*

Status: Cond (Phase-A, canonical generators, $K_0 = 7$).

2 SA2 universality

Theorem 2.1 (SA2 holds for all LS-2-satisfying Phase-A substrates with $K_0 = 7$)**.** *Under LS-2, Phase-A, $K_0 = 7$, and the canonical NFC generator class, the SA2 screened sector (Symmetry-Forcing Schema of Rank 2) is realized.*

Proof. The SA2 realization proof chain in the monograph (§28–§34) establishes SA2 conditional on LS-2 + Phase-A + PASS + $K_0 = 7$ (verified computationally for the canonical substrate). The proof uses only the algebraic properties of the $K_0 = 7$ NF2 class space (9 classes, Chevalley generators, pendant algebra structure) which are determined entirely by $K_0 = 7$ — not by any additional substrate geometry.

By Corollary 1.4: LS-2 holds for all canonical $K_0 = 7$ Phase-A substrates. Since the SA2 realization proof uses only LS-2 + Phase-A + $K_0 = 7$ (and not any substrate-specific geometric data beyond the collar alphabet), SA2 holds universally for this substrate class. □

Remark 2.2 (What remains open). Complete substrate independence — deriving LS-2 from the Part I kernel axioms alone without the three structural conditions — remains open. Paper CB establishes:

$$\text{Structural conditions (natural for Phase-A substrates)} + \text{Kernel axioms} \Rightarrow \text{LS-2} + \text{SA2}.$$

This is weaker than full independence but stronger than per-substrate computational verification. It gives a *structural* rather than merely computational foundation for the program's main conditional hypotheses.

3 Status ledger

Label	Claim	Status
CB1	Three LS-2 conditions structurally natural for Phase-A + $K_0 = 7$	COND (Phase-A)
CB2	LS-2 universal for canonical $K_0 = 7$ Phase-A substrates	COND
CB3	SA2 universal for same substrate class	COND
CB4	Full kernel-axiom-only independence	OPEN

4 Conclusion

Paper CB establishes that the suite's two principal conditional hypotheses (LS-2, SA2) hold for all canonical $K_0 = 7$ Phase-A substrates under structurally natural conditions (Phase-A + canonical generators). The computational verification of Remark 12.6 is a special case of this general structural result.

The monograph's vulnerability 1 (LS-2 substrate independence) and vulnerability 2 (SA2 substrate independence) are both addressed at the level of structural rather than full axiomatic universality.